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1 Introduction

The power to analyze relationships between large amounts of data has never been as accessible as it has been with the development of machine learning algorithms. However, there is only so much data in the world so companies that want to push what their products can do rely on large unmonitored sets of data. This can cause algorithms to be biased towards minority groups. This issue exponentiates when the usage of this technology now stretches from healthcare, recruitment, law enforcement and more. This essay will attempt to focus on salient issues brought up by the 2020 documentary "Coded Bias".

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2 Background information

The machine learning market has been developing rapidly since 2020, with the market value being projected to be around 282 billion dollars in size. Because of this rapid development, there is a larger focus on the usage of these models and the harm they can cause.

In 2020, the documentary, "Coded Bias" was released on Netflix. This documentary focuses on the research of MIT media researcher Joy Buolamwini. Her research focuses on the data collection issue plaguing machine learning models' ability to not discriminate. This research started because facial recognition algorithms were found to have significant difficulties in identifying faces of people of color and women. (DOCUMENTARY SOURCE HERE)

The documentary not only covers Buolamwini's research but also the impacts of machine learning models. One notorious example is the group Big Brother Watch in the UK which studies how the biased facial recognition systems trigger law enforcement systems with false positives and cause major stress for individuals. (BIG BROTHER WATCH SOURCE HERE)

Lastly, the documentary explores how there could be possible improvements for both the algorithms and the structures that produce the models. There is a heavy emphasis on the lack of active regulation and how the law needs to keep up with the technology being developed to prevent further issues from arising. (SOURCE HERE)

2.1 Stakeholders

There are several stakeholders in this scenario with a major power imbalance between the machine learning companies and the general public:

- Machine Learning Companies: As they decide what gets made and how it is made
- Machine Learning Engineers: As they implement the algorithms and methods of training
- Governments: As they regulate and use the algorithms for various purposes
- Software companies: As they use the algorithms to enhance their businesses
- General public: As they are the ones who are being judged by the algorithm

This power imbalance places the general public at a major disadvantage as machine learning algorithms rely on pre-existing data which is not guaranteed to be sanitized of biases. Therefore causing unjust profiling of individuals, rejections in jobs because of nationality/ethnicity, and more without any form of control as the mathematical models are a black box and companies have no incentive to reverse-engineer it.

3 Salient Issues

The documentary's main argument is that in computer systems that rely on machine learning are not accounted for in the USA's legal system at a reasonable pace. Therefore allowing companies to neglect the possible biases that the algorithms might have. This can and has been causing violations of human rights as different industries adopt this technology to handle decision-making. (BIG BROTHER WATCH SOURCE HERE and maybe other sources)

Since the original studies and the documentary, the usage of machine learning has increased across all industries (INSERT SOURCE HERE). This in turn further pushes the need for boundaries in the development, training and usage of said models by major entities like multi-national companies and governments across the world. In this essay, there will be an attempt to apply ethical frameworks to evaluate the following questions:

- What moral duties do ML companies have when producing their models?
- Who would be acting unethically for not taking responsibility when the models cause damage?
- Is it ethically feasible to enforce regulations in the black box that are ML model?

4 Technological Background

The focus of this documentary is on machine learning algorithms therefore to be able to properly analyze them there is a need to understand the following:

- Data Collection & Cleansing
  - Data collection article [Vidhya et al. Jun]
  - Data cleaning article [Lee et al. Sep]
- Types of Machine Learning [China NA]:
  - Supervised learning
  - Unsupervised learning
  - Self-supervised learning
  - Reinforced learning
  - Semi-supervised learning
- Machine Learning Algorithms (brief math overview) [Bergmann NA]

By understanding all of this, it is possible to evaluate the type of control that companies that produce and sell these algorithms have over the biases that are being baked into their software.

4.1 Data Collection and Cleansing

4.2 Machine Learning

4.3 Machine Learning Algorithms

5 Ethical Analysis

5.1 Responsibilities of ML companies

6 Beyond the superficial

A lot of DigiPen graduates tend to go out into the world and work in embedded or real-time systems that rely on ML to produce the adequate output, be it a sensor for a car, or a self-control algorithm for a plane. It is very important to understand how the usage of ML can bias the algorithms in ways we can't control as the models are trained beyond our reach.

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